# Speaker Notes

IYPT 2026 — Problem #6: The Flipo Flip

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**SLIDE 1 — Title**

- Introduce yourself and the problem

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**SLIDE 2 — Problem Statement**

- Read out the problem statement
- State the three goals of the presentation: understand the geometry, build a theoretical model connecting parameters to motion, test it with experiments

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**SLIDE 3 — Geometry: Steinmetz Solid**

- Introduce the Steinmetz solid (bicylinder): intersection of two equal-radius cylinders $R$ at right angles
- This is the geometric basis for understanding the toy

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**SLIDE 4 — Toy Construction & Moment of Inertia**

- Toy = outer sections of Steinmetz solid (top patch + mirrored bottom patch) joined by a thin square prism with rounded edges
- Both cylinders share the same rolling axis $\Rightarrow$ total moment of inertia is $mR^2$
- Because the surface comes from a Steinmetz solid, contact geometry during rolling is equivalent to a cylinder of radius $R$

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**SLIDE 5 — Phases of Motion Overview**

- Rolling motion = three repeating phases: Pure Patch Rolling, Ridge Switch, Patch Switch
- Give a brief overview before going into each

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**SLIDE 6 — Pure Patch Rolling**

- Contact point stays on one cylindrical patch $\Rightarrow$ equivalent to a cylinder of radius $R$ rolling
- Key assumption: no-slip condition, $v_{\text{CM}} = \omega R$

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**SLIDE 7 — Energy Dissipation During Pure Patch Rolling**

- Only dissipation: rolling resistance — force = $C_{rr} \cdot mg$
- Distance covered per patch = $\pi R/2$, so work done is easy to compute
- Aerodynamic drag negligible at these velocities
- Important note: rolling resistance only dissipates translational energy; rotational energy is lost through other mechanisms (e.g. switching dissipation)

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**SLIDE 8 — Velocity During Pure Patch Rolling**

- Translational velocity $u(t)$ decreases linearly with time
- $u(t) = u_0 - at$, where acceleration $a = F_{rr}/m_{\text{eff}}$
- Effective mass = physical mass + rotational inertia contribution

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**SLIDE 9 — Ridge Switch**

- Happens when contact point passes through the ridge (junction between the two cylinders)
- No additional dissipation or mechanics here — already captured in pure patch rolling
- Worth noting as a named event in the motion sequence

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**SLIDE 10 — Patch Switch (Successful)**

- Contact point hits the square edge between patches $\Rightarrow$ admissible rolling direction jumps by 90°
- If friction is sufficient: toy smoothly reorients and continues on the new patch, with some energy loss
- *[If asked: the 90° change is because the CM velocity points toward the ground before the switch, and then to the side after — that's a 90° rotation.]*

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**SLIDE 11 — Patch Switch (Failed)**

- If friction is insufficient: toy slips and yaws
- Energy dissipates rapidly $\Rightarrow$ no further flips

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**SLIDE 12 — Required Impulse for Patch Switch**

- A 90° reorientation of CM velocity from $(u, 0)$ to $(0, u)$ requires a large tangential impulse
- Compute change in translational momentum and change in angular momentum separately
- Convert angular momentum change to impulse by dividing by $R$, then sum with translational impulse
- Result: total required impulse $J_{\text{req}} = \sqrt{2}\, m_{\text{eff}}\, u$

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**SLIDE 13 — The $\Lambda$ Parameter**

- Define dimensionless $\Lambda$: ratio of available frictional impulse to required impulse $J_{\text{req}}$
- $\Lambda \ll 1$: most energy lost at each switch $\Rightarrow$ motion stops quickly
- $0 < \Lambda < 1$ in general: controls how much energy is dissipated per patch switch

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**SLIDE 14 — Full Motion Equations**

- Two sets of equations combined: pure patch rolling and patch switch dissipation
- Rolling: deceleration $k = C_{rr}\, g/2$; velocity is linear in time during each patch
- Switch: velocity after switch $= u_{\text{before}} \cdot (1 - \Lambda)$
- Assuming $\Lambda$ constant: write $u_n$ (velocity at $n$-th switch) accounting for all prior phases

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**SLIDE 15 — When Does the Toy Stop?**

- Velocity decreases linearly within each patch, then drops by factor $(1 - \Lambda)$ at each switch
- Toy can only stop during the linear decrease phase, not during a switch
- This gives the expression for $t_{\text{stop}}$

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**SLIDE 16 — Number of Flips in Closed Form**

- Toy stops at the smallest non-negative integer $n$ satisfying $u_n < k\,\Delta t$
- Solving the equality for continuous $n$ gives a closed-form expression

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**SLIDE 17 — Summary of Theoretical Model**

- Collect all derived equations on one slide
- These are the equations used to generate the theoretical predictions in the experiment plots
- Model now predicts number of patch switches before stopping

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**SLIDE 18 — Experimental Setup**

- 6 Flipo Flips with varying material, mass, and dimensions
- Equipment: laser distance meter, Tracker video analysis software, force gauge

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**SLIDE 19 — Measured Parameters**

- To feed into the model, measured separately for each toy: effective mass, patch switch time, rolling friction coefficient

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**SLIDE 20 — Force Application Locations (Setup)**

- Measured number of flips vs. initial velocity
- How the initial force is applied matters — show the possible locations

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**SLIDE 21 — Which Locations Work**

- Not all force locations initiate motion — introduce the distinction

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**SLIDE 22 — Valid Force Locations (Green vs. Other)**

- Show all tested locations; highlight in green only those that successfully started the rolling motion

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**SLIDE 23 — Experimental Procedure & Results per Toy**

- Procedure: toy placed upright, force applied to centre of surface above ridge using force gauge, initial velocity measured with Tracker, number of flips recorded
- 10 data points (initial velocity, flips) per surface type, two surface types per toy $\Rightarrow$ 20 values per toy
- Velocities tested: integer values 2–10 m/s plus 15 m/s
- Show the two graphs per toy (one per surface) with measured friction coefficients and overlay of experimental vs. theoretical predictions

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**SLIDE 24 — Model vs. Experiment: Overprediction**

- Model generally overpredicts or matches observed flips
- Likely causes: unmodelled losses (air resistance, minor deformations)
- Simplifying assumptions also contribute: $\Lambda$ treated as constant (in practice decreasing), $t_{\text{switch}}$ treated as constant (in practice increasing as velocity drops)
- No-slip assumption: toy may slip earlier than energy alone would permit

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**SLIDE 25 — Critical Velocity: Zero Flips Above a Threshold**

- Observed: above a critical velocity the toy does zero flips, while the model predicts increasing flips
- Flag this as something to explain — will address shortly

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**SLIDE 26 — Lower Friction $\Rightarrow$ Larger Error**

- Smaller $C_{rr}$ correlates with larger model error
- Explanation: low friction makes slipping more likely before energy runs out, violating the no-slip assumption more severely

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**SLIDE 27 — Trends Consistent Across All Toys**

- Confirm that overprediction, critical velocity, and friction-error correlation are seen across all 6 Flipo Flips

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**SLIDE 28 — Last Toy & Smallest Critical Velocity**

- Present results for the final Flipo Flip
- Note it has the smallest critical velocity — segue into explaining why critical velocities exist

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**SLIDE 29 — Why Does a Critical Velocity Exist?**

- Zero-flip regime: toy fails to roll at all above the threshold
- Root cause: the no-slip condition cannot hold from the very start

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**SLIDE 30 — High Velocity $\Rightarrow$ Toy Gets Launched**

- At high initial velocity, the toy cannot maintain ground contact
- It gets launched into the air instead of rolling $\Rightarrow$ no flips recorded
- The model assumes rolling from the outset, so this regime is outside its validity

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**SLIDE 31 — Bibliography**

- (No spoken content needed — just display the slide)

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**SLIDE 32 — Thank You**

- Thank the audience, invite questions

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**SLIDE 33 — (Backup / Q&A Slide)**

- No spoken text — reserve for answering questions if needed